

One Equation, Many Posets

Polynomial maps over fields as resources

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Abstract

A map implements another map when it can be placed between an input processor and an output processor. We tell the story of what follows from the single equation $f = h_{\text{out}} \circ g \circ h_{\text{in}}$. We begin with two classes of affine maps $K \rightarrow K$, then three classes of quadratic maps $\mathbb{F}_3 \rightarrow \mathbb{F}_3$, and then the first branching Hasse diagram for quadratic maps $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$. The same language leads naturally to homogeneous forms, projective zero loci, and projective rational maps.

After these examples we give the general framework for polynomial maps $K^n \rightarrow K^m$, with affine input and output processors. Invertible processors produce equivalence classes; singular processors produce a degeneration order; Cartesian powers produce asymptotic exchange rates. The familiar classification of quadratic forms—rank, radical, isotropy, discriminant, and the characteristic-two trace invariant—then appears as the first classical instance of the framework. We draw the resulting posets over finite fields, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}_2$, and $\mathbb{Q}_3$, and include larger homogeneous and cubic computations.

The companion paper solved a fiber-signature exchange theory by partition functions and Gibbs curves. Those formulas can now be evaluated on many of the polynomial classes in this paper. Independently of that formula, composition of any operational exchange rates proves $M_{ij}M_{jk} \leq M_{ik}$, so every closed conversion cycle has return at most one. We explain what cycle geometric means can measure and why this no-arbitrage property does not imply positive semidefiniteness.

The final sections are an outlook. The Euler product is an exact joint partition function of prime occupation modes for real $s > 1$, but the nontrivial zeros lie outside this ordinary Gibbs regime. Weil’s Hermitian matrix E and the directed operational matrix M presently have no proved bridge. The paper therefore offers a simple route to core concepts and a concrete research program, not a claimed connection to the Riemann hypothesis.

Guiding message. *A simple implementation rule produces core concepts. It was introduced for complexity, but it also organizes the classification and comparison of polynomial and analytic map families.*

Keywords. polynomial maps; affine equivalence; degeneration poset; quadratic forms; finite fields; exchange rate; no arbitrage; cycle mean; partition function; local zeta function.

1 Introduction

This paper begins with a very small piece of language. Let f and g be maps. We say that g implements f when

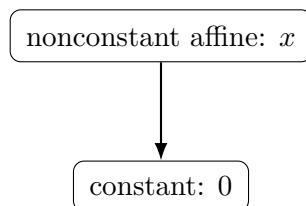
$$f = h_{\text{out}} \circ g \circ h_{\text{in}} \tag{1}$$

for allowed input and output processors. Thus one call to g , together with preprocessing and postprocessing, realizes f . This was the starting point of the finite-map complexity theory in the companion paper [1]. Here we keep the sentence and change the objects: the maps are polynomials over a field, and the main processors are affine.

We will not begin with the most general definition. The first examples are small enough to see completely.

First example: affine maps of a line

Consider degree-at-most-one maps $K \rightarrow K$, with affine input and output processors. Every nonconstant affine map is equivalent to x , and every constant map is equivalent to 0. The map x implements 0 by a constant output processor. A constant cannot implement x . We have already obtained a two-node Hasse diagram:



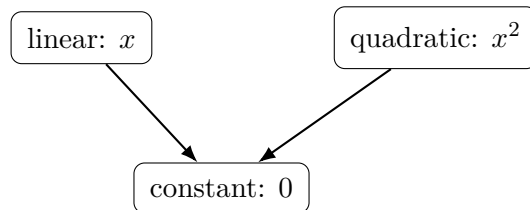
The two core concepts—equivalence class and degeneration order—came from the same implementation rule.

Second example: quadratic maps $\mathbb{F}_3 \rightarrow \mathbb{F}_3$

Every degree-at-most-two polynomial over $\mathbb{F}_3$ is uniquely a function of the form $ax^2 + bx + c$. Invertible affine changes of input and output give three classes:

$$0, \quad x, \quad x^2.$$

Their class sizes are 3, 6, 18, respectively, accounting for all $3^3 = 27$ quadratic polynomial functions of one variable. Both nonconstant classes implement a constant. Neither implements the other: a nonconstant affine substitution preserves the degree of x^2 , while a singular substitution is constant. The poset branches:



This is the first warning that “more complicated” is not automatically a linear ranking. The linear and quadratic classes lose different structure.

Third example: quadratic maps $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$

Now take

$$Q(x, y) = ax^2 + bxy + cy^2 + dx + ey + f.$$

There are $3^6 = 729$ coefficient vectors, but only six affine-equivalence classes:

$$0, \quad x, \quad x^2, \quad x^2 + y, \quad xy, \quad x^2 + y^2.$$

The last two are the split and anisotropic nonsingular quadratic forms. The split form has an isotropic line and can lose its quadratic part along that line, producing x . The anisotropic form has no such line, so it reaches x^2 but not x . Figure 1 later displays the full six-node Hasse diagram with all class sizes.

This example already touches classical quadratic-form theory. The framework has not replaced the classical invariants; it has explained why they appear. Rank, the radical, and isotropy are precisely the information that survives or disappears under singular affine processors.

Why go projective?

Translations mix homogeneous degrees. If we want to study the leading part of a polynomial, or what its fibers do at infinity, the natural next step is to keep only homogeneous maps and use linear processors. A homogeneous scalar polynomial $Q \in \text{Sym}^d(V^*)$ scales by $Q(\lambda v) = \lambda^d Q(v)$, so its zero set descends to projective space $\mathbf{P}(V)$.

There is a small but important distinction. A scalar form on V does *not* give an interesting map $\mathbf{P}(V) \rightarrow \mathbf{P}(K)$, because $\mathbf{P}(K) = \mathbf{P}^0$ is one point. It is naturally a section of $\mathcal{O}(d)$, or equivalently a projective hypersurface $Q = 0$. A tuple

$$F = (F_0, \dots, F_r), \quad F_i \in \text{Sym}^d(V^*),$$

does give a rational map

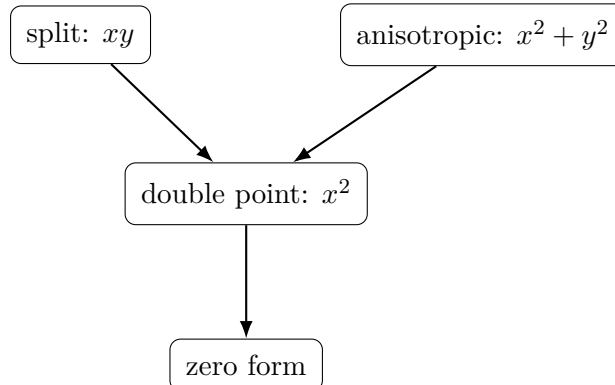
$$\mathbf{P}(V) \dashrightarrow \mathbf{P}^r, \quad [v] \mapsto [F_0(v) : \dots : F_r(v)].$$

It is a total projective morphism exactly when the forms have no common projective zero. Thus scalar forms and tuples belong in the same tensor framework, but their geometric readings are different.

Over $\mathbb{F}_3$, binary quadratic forms have four classes:

$$0, \quad x^2, \quad xy, \quad x^2 + y^2.$$

Their orbit sizes are respectively 1, 8, 12, 6, summing to the $3^3 = 27$ formal binary quadratics. The last two are different degree-two divisors on $\mathbf{P}^1$: split over $\mathbb{F}_3$, or conjugate over $\mathbb{F}_9$. Their degeneration diagram is



Pairs of binary quadratics give rational self-maps $\mathbf{P}^1 \dashrightarrow \mathbf{P}^1$. Their seven classes and eight Hasse covers appear in Figure 6; the base-point-free nodes are genuine quadratic morphisms. Ternary quadratic forms similarly give projective conics in $\mathbf{P}^2$, with five classes. These examples will be drawn in Section 6.

We now return to the general story. Depending on the processor class, the same implementation equation appears as a reduction in complexity theory, a simulation in information theory, a restriction of tensors, or a left–right action in singularity theory. The point is to keep the equation fixed and vary the processors deliberately.

The next continuation is where the companion paper [1] enters. For finite maps with a fiber-preserving processor convention, the asymptotic exchange rate is computed exactly from the partition functions

$$Z_f(\beta) = \sum_y |f^{-1}(y)|^\beta.$$

Many polynomial classes have the same fiber signature, while others do not, so this produces a second numerical geometry on the class poset. It is a useful observer, but it must not be confused with the block-affine exchange rate. The two processor classes are different.

The paper has three levels of certainty.

1. The processor preorder, the quadratic normal forms, the finite-field class counts, and the no-arbitrage theorem are proved or reduced to standard quadratic-form calculations.
2. The larger homogeneous and cubic diagrams, and the displayed exchange matrices, are reproducible finite computations.
3. The possible relation among exchange matrices, local zeta profiles, Weil positivity, and the Riemann hypothesis is exploratory. We state precisely what is known and what is not connected.

This separation is part of the contribution. It prevents a suggestive common vocabulary—resources, partition functions, spectra, exchange matrices—from being mistaken for a theorem that has not yet been found.

2 Processors, equivalence, and order

Fix a field K , dimensions n, m , and a degree bound d . Write

$$\text{Pol}_d(K^n, K^m)$$

for polynomial maps of total degree at most d . An affine input processor and output processor have the form

$$a(x) = Ax + t, \quad b(y) = By + u,$$

with $A \in \text{Mat}_{n \times n}(K)$ and $B \in \text{Mat}_{m \times m}(K)$.

Definition 2.1 (Affine implementation). For $f, g \in \text{Pol}_d(K^n, K^m)$, say that g implements f , and write $f \preceq_{\text{aff}} g$, when

$$f = b \circ g \circ a \tag{2}$$

for affine a and b . The matrices A and B may be singular, including zero.

The convention is that the resource is on the right of $\preceq_{\text{aff}}$: an arrow in a Hasse diagram points from g down to a map f that it implements.

Proposition 2.2. *Affine implementation is a preorder.*

Proof. Identity processors prove reflexivity. If $f = b_1 \circ g \circ a_1$ and $g = b_2 \circ h \circ a_2$, then

$$f = (b_1 \circ b_2) \circ h \circ (a_2 \circ a_1),$$

and affine maps are closed under composition. $\square$

Invertible processors define the affine left–right equivalence relation

$$f \sim_{\text{aff}} g \iff f = b \circ g \circ a, \quad a \in \text{Aff}_n(K), \quad b \in \text{Aff}_m(K) \text{ invertible.} \quad (3)$$

This is the finite-dimensional affine version of the left–right equivalence used for map germs and singularities; see, for example, [4]. The set $\text{Pol}_d(K^n, K^m)$ is a representation space for the invertible affine group, and its equivalence classes are group orbits.

Singular processors induce a directed relation on those orbits. In every quadratic diagram below this relation is antisymmetric. In general it need not be: if two different invertible orbits are mutually reachable using singular processors, one must contract the strongly connected component. The quotient by mutual reachability is always a poset.

Remark 2.3 (Exact reachability versus orbit closure). Our arrows solve the exact equation (2) with possibly singular processors. This is related to, but not automatically the same as, containment of algebraic-group orbit closures. Orbit-closure theory allows limits of invertible transformations and is governed by results such as [5]; an exact singular factorization is a more specific certificate. The two orders should not be identified without proof.

Remark 2.4 (Bounded-degree nonlinear processors). If input processors have degree at most two, a one-step relation is generally not transitive because two quadratic substitutions may compose to degree four. The correct object is then the transitive closure of allowed steps, followed by quotienting mutual reachability. This distinction explains the large collapse in the cubic $\mathbb{F}_3$ example of Section 6.

The implementation order comes with immediate monotones: polynomial degree, the dimension of the affine span of the image, coordinate rank for homogeneous tensors, and appropriate ranks of derivatives or polar forms cannot increase under the corresponding processors. These monotones help prove that proposed arrows are Hasse covers and that omitted arrows are impossible.

3 Cartesian powers and the operational rate

For $f : K^n \rightarrow K^m$, let

$$f^{\times r} : K^{nr} \longrightarrow K^{mr}, \quad f^{\times r}(x_1, \dots, x_r) = (f(x_1), \dots, f(x_r)).$$

The block processor is allowed to mix the r copies. Define

$$k_{g \rightarrow f}(r) = \max\{k : f^{\times k} \preceq_{\text{aff}} g^{\times r}\}.$$

Block-diagonal composition gives $k(r+s) \geq k(r) + k(s)$, so Fekete’s lemma gives the limit

$$C_{\text{aff}}(g \rightarrow f) = \lim_{r \rightarrow \infty} \frac{k_{g \rightarrow f}(r)}{r} = \sup_{r \geq 1} \frac{k_{g \rightarrow f}(r)}{r}, \quad (4)$$

with infinite values permitted for degenerate resources.

The same construction works for any processor family that is closed under composition and block sum. The following consequence is the fundamental “no-arbitrage” statement.

Theorem 3.1 (Composition and no arbitrage). *Let $C(i \rightarrow j)$ be operational asymptotic rates in such a resource theory. Then*

$$C(i \rightarrow j)C(j \rightarrow k) \leq C(i \rightarrow k). \quad (5)$$

For normalized nondegenerate resources with $C(i \rightarrow i) = 1$, every directed cycle $\mathcal{C} = (i_0, i_1, \dots, i_\ell = i_0)$ satisfies

$$R(\mathcal{C}) := \prod_{r=0}^{\ell-1} C(i_r \rightarrow i_{r+1}) \leq 1. \quad (6)$$

Proof. Choose block conversions with rates arbitrarily close to $C(i \rightarrow j)$ and $C(j \rightarrow k)$, clear the block denominators, and compose them. This realizes any rate strictly below their product from i to k , proving (5). Iterating around a cycle gives $R(\mathcal{C}) \leq C(i_0 \rightarrow i_0) = 1$. $\square$

For a finite collection put $M_{ij} = C(i \rightarrow j)$. When all entries are positive, the logarithmic costs

$$a_{ij} = -\log M_{ij} \quad (7)$$

satisfy the directed triangle inequality

$$a_{ik} \leq a_{ij} + a_{jk}$$

and every directed cycle has nonnegative total cost. The costs need not be nonnegative edge by edge because one resource may produce more than one copy of another; it is the cycle sum that cannot be negative.

3.1 Why no arbitrage does not imply positive semidefiniteness

The matrix M is directed and normally nonsymmetric, so positive semidefiniteness is not even an intrinsic property of it. More strongly, entrywise positivity, unit diagonal, and all cycle inequalities do not imply positive semidefiniteness even after symmetry is imposed. For example,

$$S = \begin{pmatrix} 1 & 0.9 & 0.9 \\ 0.9 & 1 & 0.1 \\ 0.9 & 0.1 & 1 \end{pmatrix}.$$

Every cycle product is at most one, but

$$\det S = 1 + 2(0.9)(0.9)(0.1) - 0.9^2 - 0.9^2 - 0.1^2 = -0.468 < 0.$$

Thus the no-arbitrage cone and the positive-semidefinite cone are genuinely different geometries.

4 Quadratic maps $K^2 \rightarrow K$

Consider

$$Q(x, y) = ax^2 + bxy + cy^2 + dx + ey + f. \quad (8)$$

For $\text{char } K \neq 2$, write

$$Q(z) = z^\top S z + \ell^\top z + c$$

with S symmetric. Translation by t replaces the linear part by $\ell + 2St$. Hence the component of ℓ in the image of S can be removed. What remains is a linear functional on the radical of S . This gives four preliminary types:

$$\text{constant}, \quad x, \quad x^2, \quad x^2 + y,$$

followed by the similarity classes of nonsingular binary quadratic forms.

The form $x^2 + y$ is called parabolic here. Its linear term lives along the radical of its quadratic part, and no translation removes it. A nonsingular binary form has no radical, so every linear term can be removed.

Under a singular input matrix, a nondegenerate form always restricts to a rank-one form x^2 . It restricts to a nonconstant linear map exactly when it has an isotropic direction: choosing that direction kills the quadratic term while a translation in a transverse direction creates the linear term. This one observation determines the split/anisotropic difference in every diagram below.

Theorem 4.1 (Odd finite fields). *For every odd prime power q , affine equivalence on degree-at-most-two maps $\mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ has six classes:*

$$1, \quad x, \quad x^2, \quad x^2 + y, \quad xy, \quad x^2 - \nu y^2,$$

where ν is a nonsquare. Their numbers of coefficient vectors are

<i>type</i>	<i>class size</i>
<i>constant</i>	q
<i>linear</i>	$q(q^2 - 1)$
<i>rank-one quadratic</i>	$q^2(q^2 - 1)$
<i>parabolic</i>	$q^2(q - 1)(q^2 - 1)$
<i>split quadratic</i>	$q^4(q^2 - 1)/2$
<i>anisotropic quadratic</i>	$q^4(q - 1)^2/2$

The Hasse covers are

$$\begin{aligned} [x^2 + y] &\longrightarrow [x], [x^2], \\ [xy] &\longrightarrow [x], [x^2], \\ [x^2 - \nu y^2] &\longrightarrow [x^2], \\ [x], [x^2] &\longrightarrow [1]. \end{aligned}$$

Proof sketch. The preceding translation argument gives the four singular types. The usual Witt classification of nonsingular binary forms over $\mathbb{F}_q$ gives one split and one anisotropic similarity class; see [6, 7]. The arrows follow from rank-one restrictions and isotropy. Orbit-stabilizer counts give the displayed sizes, which sum to q^6 . $\square$

Figure 1 is the case $q = 3$. Its six class sizes are 3, 24, 72, 144, 324, 162, summing to $3^6 = 729$.

4.1 Characteristic two

In characteristic two the polar bilinear form no longer determines the quadratic form, and square terms behave additively. Over $\mathbb{F}_2$, polynomial functions satisfy $x^2 = x$ and $y^2 = y$, leaving only three classes: constant, balanced linear, and weight-one-or-three, represented by $1, x, xy$.

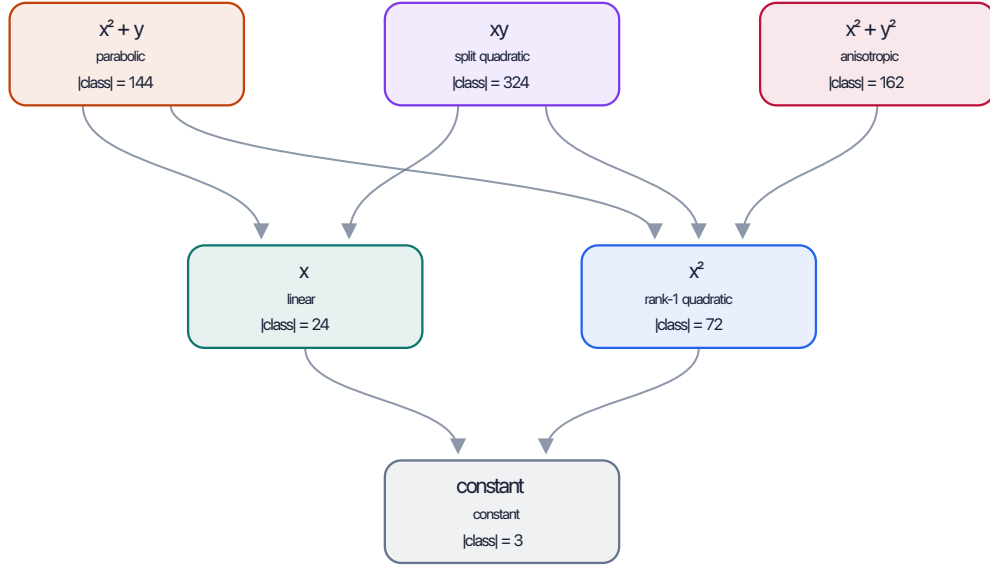


Figure 1: The affine-processor Hasse diagram for quadratic maps $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$: six classes and seven covers. Singular affine input and output processors are allowed. Arrows point from a resource map to an immediately implementable degeneration; class sizes count coefficient vectors.

For even $q \geq 4$, there are seven classes. In addition to the constant, linear, pure-square, parabolic, split, and anisotropic classes, there is an aligned-linear class $x^2 + x$. With $\delta \in \mathbb{F}_q$ satisfying $\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(\delta) = 1$, the nonsingular representatives are xy and $x^2 + xy + \delta y^2$. The class sizes are

type	class size
constant	q
linear	$q(q^2 - 1)$
pure square	$q(q^2 - 1)$
square plus aligned linear	$q(q - 1)(q^2 - 1)$
parabolic	$q^2(q - 1)(q^2 - 1)$
split quadratic	$q^4(q^2 - 1)/2$
anisotropic quadratic	$q^4(q - 1)^2/2$.

Again the sum is q^6 . The trace/Arf invariant replaces the discriminant test. The aligned-linear node is incomparable with the pure-square and ordinary linear nodes, although all three degenerate to a constant.

5 Changing the field

The processor definition does not change with the field; only the quadratic similarity classes and isotropic directions change.

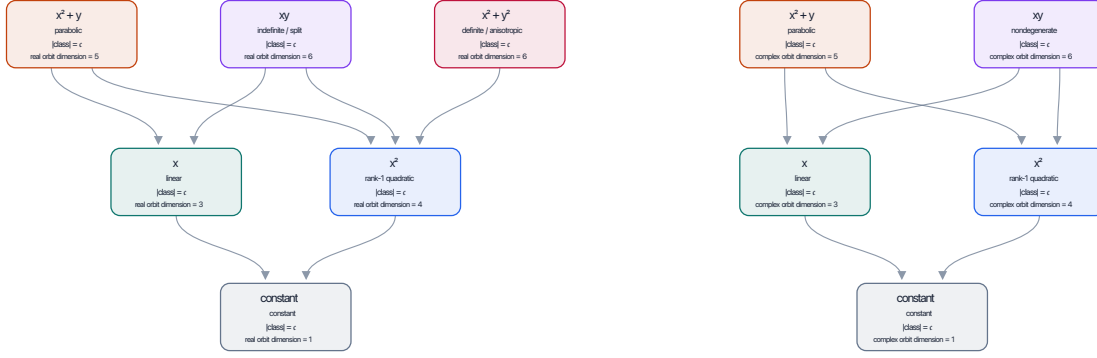


Figure 2: The same affine implementation convention over $\mathbb{R}$ and $\mathbb{C}$. The real diagram has six classes and seven covers; the complex diagram has five classes and six covers. Singular affine input and output processors are allowed. The field changes the nonsingular quadratic-form classes, not the definition of the order.

5.1 The real and complex numbers

Over $\mathbb{C}$, every nonsingular binary quadratic form is similar to xy , so the diagram has five nodes. Over $\mathbb{R}$, output rescaling identifies positive and negative definite forms, leaving two nonsingular classes: the indefinite form xy and the definite form $x^2 + y^2$. The real diagram therefore has six nodes. The definite class has no arrow to x , exactly because it has no real isotropic direction.

All nonempty real or complex orbits have continuum cardinality. Orbit dimension is the useful label: constant, linear, rank-one, parabolic, and open nonsingular strata have dimensions 1, 3, 4, 5, 6, respectively, over the base field. These are real dimensions for $\mathbb{R}$ and complex dimensions for $\mathbb{C}$.

5.2 Local fields

For a non-Archimedean local field, the square-class group creates several nonsingular binary forms. Over $\mathbb{Q}_3$, the representatives $d \in \{1, -1, 3, -3\}$ in

$$x^2 + dy^2$$

give one split class $d = -1$ and three pairwise-incomparable anisotropic classes. Together with the four singular types, this gives eight nodes.

Over $\mathbb{Q}_2$, the square classes

$$\{1, -1, 2, -2, 5, -5, 10, -10\}$$

give one split and seven anisotropic classes, hence twelve nodes. Every anisotropic node reaches x^2 but not x , and the seven are pairwise incomparable in the one-shot affine poset. Figure 4 displays the result.

The orbit dimension labels are again 1, 3, 4, 5, 6 in the six-dimensional coefficient space. Cardinality cannot distinguish the orbits because every class has continuum size.

6 Beyond scalar quadratic maps

The same construction applies to homogeneous tensors

$$F \in \text{Hom}(\text{Sym}^d V, W), \quad G = B \circ F \circ A.$$

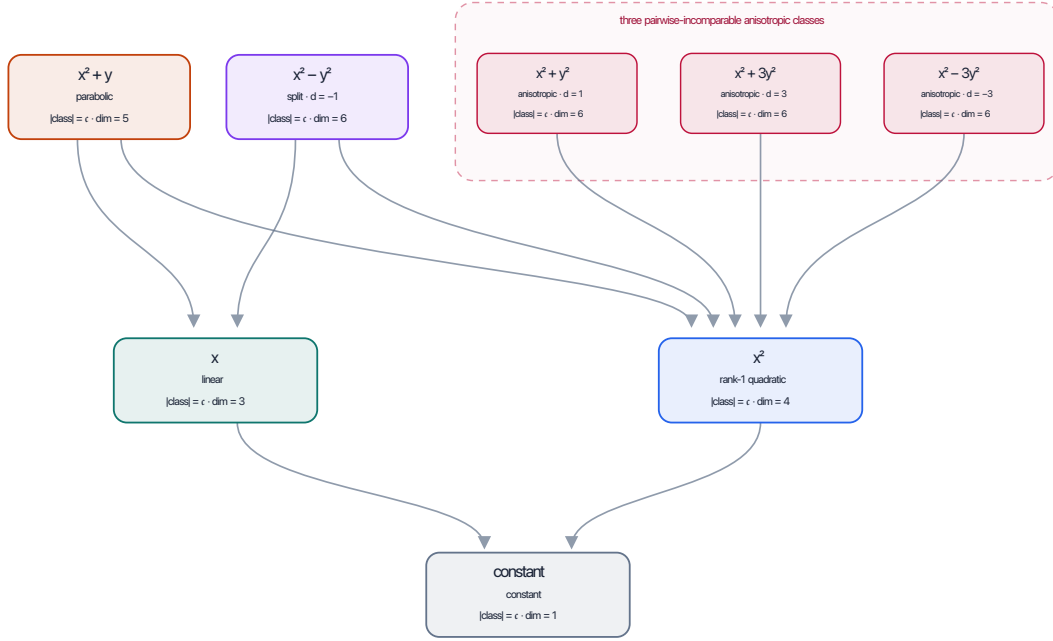


Figure 3: Quadratic maps $\mathbb{Q}_3^2 \rightarrow \mathbb{Q}_3$ with singular affine processors: eight classes and nine covers. The three anisotropic square classes are incomparable and have no arrow to the linear node.

Here translations are removed to preserve homogeneity. Invertible $A \in \text{GL}(V)$, $B \in \text{GL}(W)$ define equivalence; arbitrary linear maps define degeneration.

Exhaustive calculations over $\mathbb{F}_3$ give the following atlas.

tensor family	total tensors	classes	Hasse covers
ternary quadratic forms	3^6	5	5
quadratic $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$	3^6	7	8
quadratic $\mathbb{F}_3^3 \rightarrow \mathbb{F}_3^3$	3^{18}	50	210
ternary cubic forms	3^{10}	26	66
cubic $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$	3^8	19	32

The quadratic 3-input, 3-output calculation quotients output changes first, replacing coordinate tuples by subspaces of the six-dimensional space of ternary quadrics. Only 45,256 subspaces of dimension at most three need be enumerated instead of all 3^{18} tensors.

The first row of the table is the projective-conic example promised in the introduction. A ternary quadratic form is a section on $\mathbf{P}^2$; its zero locus is the whole plane for the zero form, a double line, two lines, an anisotropic rank-two conic, or a nonsingular conic. Singular linear input processors contract these projective configurations exactly as shown in Figure 5.

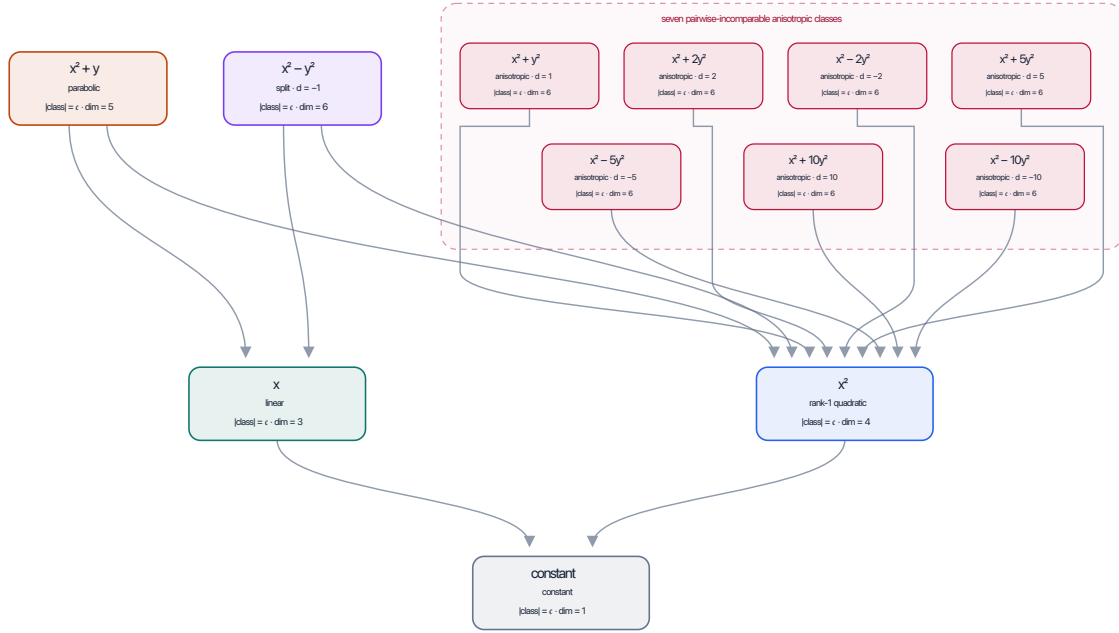


Figure 4: Quadratic maps $\mathbb{Q}_2^2 \rightarrow \mathbb{Q}_2$ with singular affine processors: twelve classes and thirteen covers. Seven anisotropic classes sit above the rank-one node. Their one-shot incomparability is exact; their block-affine exchange rates are largely unknown.

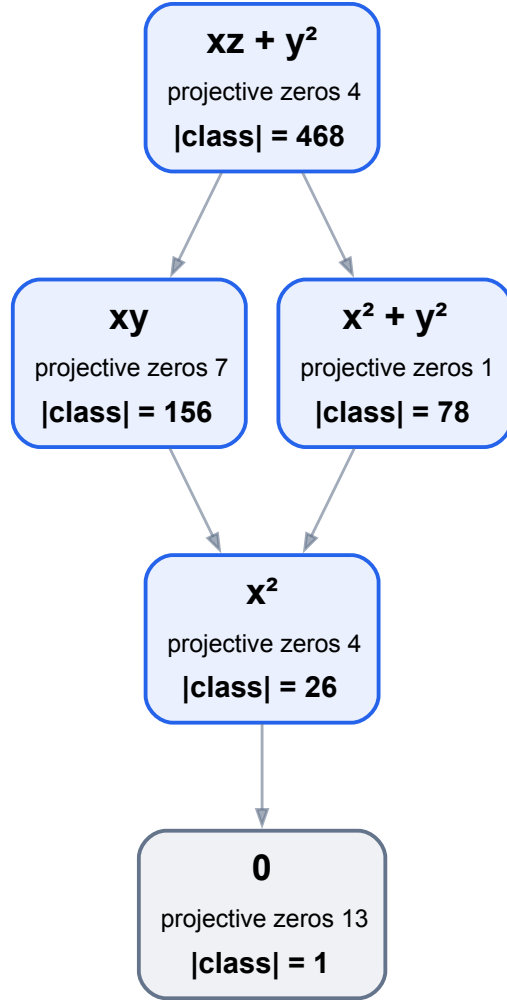


Figure 5: Ternary quadratic forms over $\mathbb{F}_3$, read as sections of $\mathcal{O}(2)$ on $\mathbf{P}^2$: five classes and five covers under possibly singular linear input processors and scalar output processors. The tensor classes are also five types of projective conic, including degenerate types.

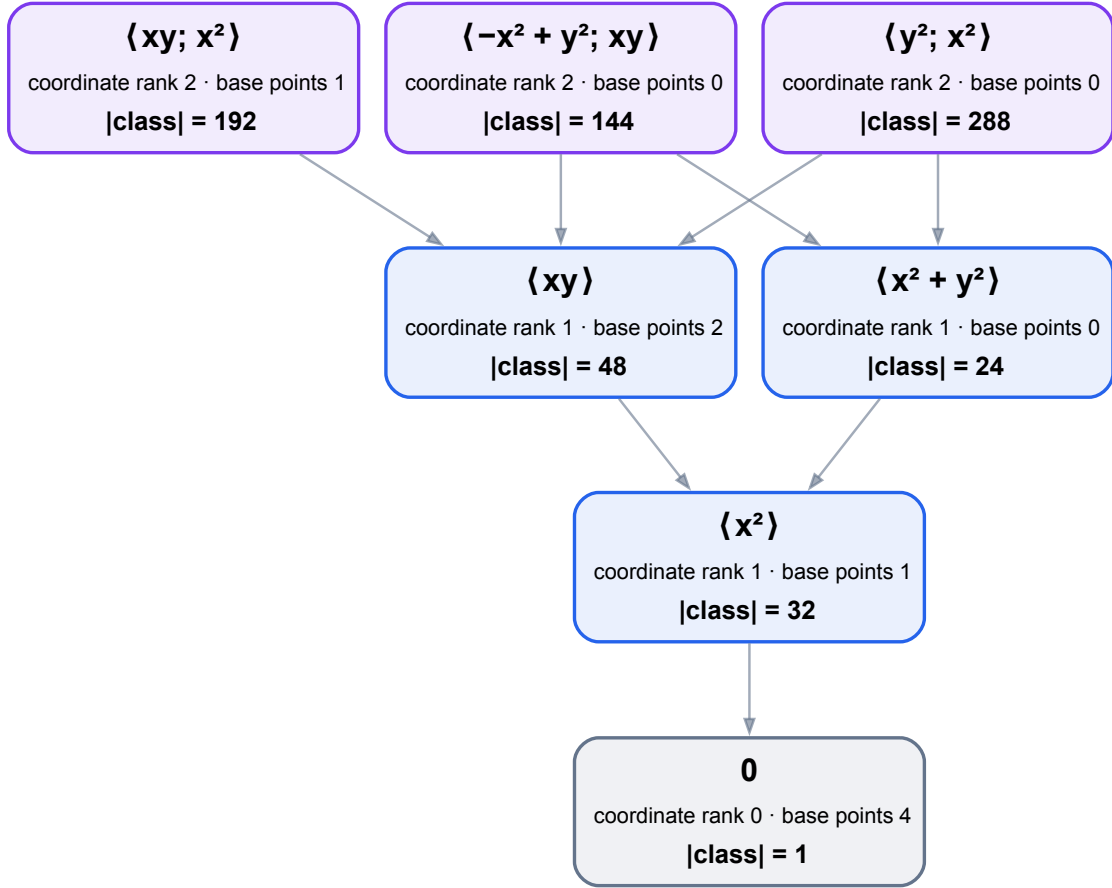


Figure 6: Seven homogeneous quadratic tensor classes and eight covers for $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$, with possibly singular linear input and output processors. This example will also carry a fiber-signature exchange matrix in Section 7.

For degree-at-most-three functions $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$, affine input and output processors give 14 classes and 22 Hasse covers. If quadratic input processors are allowed and the generated preorder is used, the 6,561 functions collapse to three mutual-reachability classes: surjective, two-valued with fibers $6 + 3$, and constant.

Over $\mathbb{F}_8$, the functional identities responsible for this collapse no longer hold. Among $8^{10} = 1,073,741,824$ cubic functions, the quadratic-input generated preorder has 110 classes. The exact Hasse diagram has 23 boxes after grouping order-theoretic twin antichains. This contrast is a warning: bounded-degree substitutions interact strongly with the arithmetic of the field and with polynomial identities as functions.

7 The fiber-signature observer

For a finite map $f : X \rightarrow Y$, discard the labels and retain the nonzero fiber sizes

$$\sigma(f) = (|f^{-1}(y_1)|, \dots, |f^{-1}(y_r)|).$$

The companion paper [1] studies a processor convention that injects each target fiber into an assigned source fiber. For that resource theory the exchange rate is

$$C_{\text{sig}}(g \rightarrow f) = \inf_{\beta \in [0, \infty]} \frac{\log Z_g(\beta)}{\log Z_f(\beta)}, \quad Z_f(\beta) = \sum_y |f^{-1}(y)|^\beta. \quad (9)$$

Cartesian products multiply the partition functions, and the functions $Z_f(\beta)$ form the relevant asymptotic spectrum. Equivalently, (9) is homothetic containment of Gibbs energy–entropy regions.

Remark 7.1 (No silent identification with the affine rate). The affine and signature processors are not the same. An affine input may be singular, while the signature model preserves fibers by injections; conversely, the signature model forgets all algebraic restrictions. Therefore C_{sig} is used here as a computable observer attached to a polynomial map, not as a proved value or bound for C_{aff} . Any comparison theorem between the two rates requires an explicit functor between the processor categories.

7.1 A six-node matrix

Remove the zero node from Figure 6 and order the other nodes as

$$\begin{aligned} A_1 &= \langle x^2 \rangle, & A_2 &= \langle xy \rangle, & A_3 &= \langle x^2 + y^2 \rangle, \\ A_4 &= \langle xy; x^2 \rangle, & A_5 &= \langle y^2; x^2 \rangle, & A_6 &= \langle -x^2 + y^2; xy \rangle. \end{aligned}$$

Their signatures are

$$(6, 3), (5, 2, 2), (4, 4, 1), (3, 2, 2, 2), (4, 2, 2, 1), (2, 2, 2, 2, 1).$$

Thus the partition profiles for all seven nodes in Figure 6, including the zero node A_0 , are

$$\begin{aligned} \mathcal{Z}_0(s) &= 9^s, \\ \mathcal{Z}_1(s) &= 6^s + 3^s, \\ \mathcal{Z}_2(s) &= 5^s + 2 \cdot 2^s, \\ \mathcal{Z}_3(s) &= 2 \cdot 4^s + 1, \\ \mathcal{Z}_4(s) &= 3^s + 3 \cdot 2^s, \\ \mathcal{Z}_5(s) &= 4^s + 2 \cdot 2^s + 1, \\ \mathcal{Z}_6(s) &= 4 \cdot 2^s + 1. \end{aligned}$$

Equation (9) gives the matrix whose rows are resources and columns are targets:

$$M_{\text{sig}} = \begin{pmatrix} 1 & .630930 & .630930 & .500000 & .500000 & .430677 \\ .897378 & 1 & .996600 & .792481 & .792481 & .682606 \\ .773706 & .861353 & 1 & .792481 & .792481 & .682606 \\ .613147 & .682606 & .755027 & 1 & .792481 & .861353 \\ .773706 & .861353 & .904077 & .985905 & 1 & .861353 \\ .386853 & .430677 & .500000 & .630930 & .500000 & 1 \end{pmatrix}.$$

Every entry is positive and every two-way product is at most one. More strongly, the full composition inequalities $M_{ij}M_{jk} \leq M_{ik}$ hold, as guaranteed by the operational signature theory.

The matrix is not symmetric. Its ordinary eigenvalues are approximately

$$4.55109, 0.70981, 0.35523, 0.21839, 0.08274 \pm 0.00412i.$$

The leading eigenvalue exceeding one does not represent arbitrage: ordinary matrix multiplication sums over many alternative paths, whereas a conversion chooses and composes a path. The relevant algebra for best-path conversion is max-times, not ordinary linear algebra.

8 Cycle means: what they measure and what they do not

For a directed simple cycle $\mathcal{C}$, define its total return and per-step retention by

$$R(\mathcal{C}) = \prod_{(i,j) \in \mathcal{C}} M_{ij}, \quad r(\mathcal{C}) = R(\mathcal{C})^{1/|\mathcal{C}|}.$$

The no-arbitrage theorem gives $r(\mathcal{C}) \leq 1$. In logarithmic costs, $-\log r(\mathcal{C})$ is the mean loss per conversion. Maximizing this geometric mean is the maximum-mean-cycle problem, or the max-times spectral radius after trivial diagonal loops are removed. It is computable by the additive cycle-mean methods associated with [9].

For the six-node matrix above, the best simple cycle of each length is:

length	$r(\mathcal{C})$	$R(\mathcal{C})$	maximizing cycle
2	0.926512	0.858424	$A_2 \rightarrow A_3 \rightarrow A_2$
3	0.879489	0.680285	$A_2 \rightarrow A_3 \rightarrow A_5 \rightarrow A_2$
4	0.856880	0.539113	$A_2 \rightarrow A_3 \rightarrow A_4 \rightarrow A_5 \rightarrow A_2$
5	0.789265	0.306278	$A_1 \rightarrow A_3 \rightarrow A_4 \rightarrow A_5 \rightarrow A_2 \rightarrow A_1$
6	0.741674	0.166448	$A_1 \rightarrow A_3 \rightarrow A_5 \rightarrow A_6 \rightarrow A_4 \rightarrow A_2 \rightarrow A_1$

These numbers say how nearly reversible the best closed portfolios are at a fixed number of distinct classes. They do not recover the Hasse order: a one-shot incomparable pair may still have positive block exchange rates.

For the seven pairwise-incomparable anisotropic $\mathbb{Q}_2$ classes, the true affine matrix is not known. A finite diagnostic reduces the forms modulo 2^{10} , forgets the affine structure, and computes their residue fiber-signature matrix. It has nontrivial cycles with mean exactly one among classes whose residue signatures coincide; its best full seven-class cycle has

$$r_7 = 0.9923402268, \quad R_7 = 0.9475980924.$$

This reveals that the residue observer is too coarse to retain the seven square classes. It is not evidence that the exact affine resources are reversible.

Thus cycle means have three defensible uses:

1. checking numerical no-arbitrage and composition consistency;
2. quantifying round-trip loss inside a chosen rate model;
3. detecting when an observer collapses algebraically distinct classes.

They do not by themselves supply a scalar complexity, a Hasse order, a positive-semidefinite form, or an affine coding theorem.

9 The unresolved 2-adic affine rates

Let

$$q_d(x, y) = x^2 + dy^2.$$

For the pair $q_1 = x^2 + y^2$ and $q_2 = x^2 + 2y^2$, two copies of either form implement one copy of the other: restrict two source blocks to $(x, 0)$ and $(y, 0)$, then combine the outputs. Therefore

$$C_{\text{aff}}(q_1 \rightarrow q_2), C_{\text{aff}}(q_2 \rightarrow q_1) \geq \frac{1}{2}.$$

An equal-block n -to- n conversion is impossible for distinct determinant square classes, but this does not exclude rates approaching one. The current rigorous bounds remain

$$\frac{1}{2} \leq C_{\text{aff}}(q_1 \rightarrow q_2) \leq 1, \quad \frac{1}{2} \leq C_{\text{aff}}(q_2 \rightarrow q_1) \leq 1.$$

No integral-lattice 3-to-2 identity exists modulo four, and a natural shared-summand ansatz also fails. Rational processors with denominators and general cross-copy mixing remain open.

There is a separate local-zeta diagnostic. With normalized Haar measure on $\mathbb{Z}_p^n$, define

$$Z_{f,p}(s) = \int_{\mathbb{Z}_p^n} |f(x)|_p^s dx, \quad C_\zeta(f \rightarrow g) = \inf_{s>0} \frac{-\log Z_{f,p}(s)}{-\log Z_{g,p}(s)}.$$

For $t = 2^{-s}$,

$$Z_{x^2,2}(s) = \frac{1}{2-t^2}, \quad Z_{x^2-y^2,2}(s) = \frac{t^2-2t+2}{(2-t)^2},$$

giving

$$C_\zeta(x^2 \rightarrow x^2 - y^2) = 0.9397027875 \dots$$

Igusa's rationality theory explains why these local profiles are rational in p^{-s} [10]. But they depend on the chosen lattice and are invariant under $\text{GL}_n(\mathbb{Z}_p)$, not unrestricted $\text{GL}_n(\mathbb{Q}_p)$. They are prices in a local profile model, not the operational affine rates above.

10 Prime modes and the zeta partition function

For $\Re s > 1$, unique factorization gives the exact identities

$$\zeta(s) = \sum_{n \geq 1} e^{-s \log n} = \prod_p \left(\sum_{k \geq 0} e^{-sk \log p} \right) = \prod_p \frac{1}{1 - p^{-s}}. \quad (10)$$

This is a genuine Gibbs interpretation for real $s = \beta > 1$. An “atom” at a prime p is an independent bosonic mode with occupation number $k \geq 0$ and energy

$$E_p(k) = k \log p.$$

Its partition function is

$$Z_p(\beta) = \frac{1}{1 - p^{-\beta}},$$

and the joint energy of an occupation vector $(k_p)_p$ is $\sum_p k_p \log p = \log n$. This elementary oscillator picture is compatible with, but much less structured than, the Bost–Connes quantum statistical system whose partition function is $\zeta(\beta)$ [11].

For one prime mode, the geometric occupation law gives

$$\bar{k}_p(\beta) = \frac{1}{p^\beta - 1}, \quad U_p(\beta) = \frac{\log p}{p^\beta - 1}, \quad S_p(\beta) = -\log(1 - p^{-\beta}) + \beta U_p(\beta).$$

Figure 7 compares the first six primes. At a fixed energy, a larger p has fewer accessible occupation levels and therefore smaller entropy. If $u = U_p / \log p = \bar{k}_p$, however, every curve reduces to the same universal bosonic relation

$$S = (1 + u) \log(1 + u) - u \log u.$$

Each individual prime mode is normalizable for every $\beta > 0$, although their infinite product is normalizable only for $\beta > 1$. No individual mode has a negative-temperature branch because its spectrum is unbounded above.

For the normalized Gibbs law, in units with $k_B = 1$, the mean energy and entropy are

$$U(\beta) = -\frac{d}{d\beta} \log \zeta(\beta), \quad S(\beta) = \log \zeta(\beta) + \beta U(\beta).$$

Figure 8 plots this parametric entropy–energy curve. It has only the branch $\beta > 1$: as $\beta \rightarrow \infty$, both U and S tend to zero, while both diverge as $\beta \downarrow 1$.

If the exchange-profile formula is applied formally to two individual prime modes, then

$$C_{\text{prof}}(\text{Atom}(p) \rightarrow \text{Atom}(q)) = \inf_{\beta > 0} \frac{-\log(1 - p^{-\beta})}{-\log(1 - q^{-\beta})} = \begin{cases} 1, & p \leq q, \\ 0, & p > q. \end{cases} \quad (11)$$

Indeed, for $p < q$ the ratio is at least one and tends to one as $\beta \downarrow 0$; for $p > q$ it tends to zero as $\beta \rightarrow \infty$. This degenerate answer is informative: over the full temperature range, unequal infinite ladders do not have matching low- and high-temperature asymptotics. A nontrivial comparison needs a temperature window, a work offset, or a different operational model. Equation (11) is therefore a profile calculation, not a new coding theorem for primes.

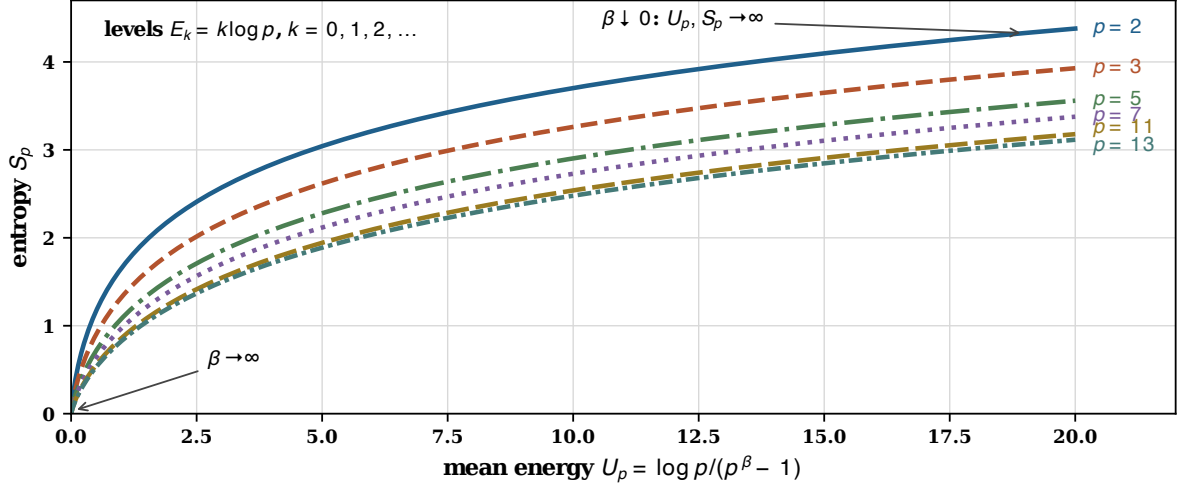


Figure 7: Entropy–energy curves for the first six prime modes $p = 2, 3, 5, 7, 11, 13$, each with levels $E_k = k \log p$. The curves are distinct in physical energy U_p , but coincide after the horizontal rescaling $U_p \mapsto U_p / \log p$. Their canonical domain is $\beta > 0$; negative temperatures do not define Gibbs states.

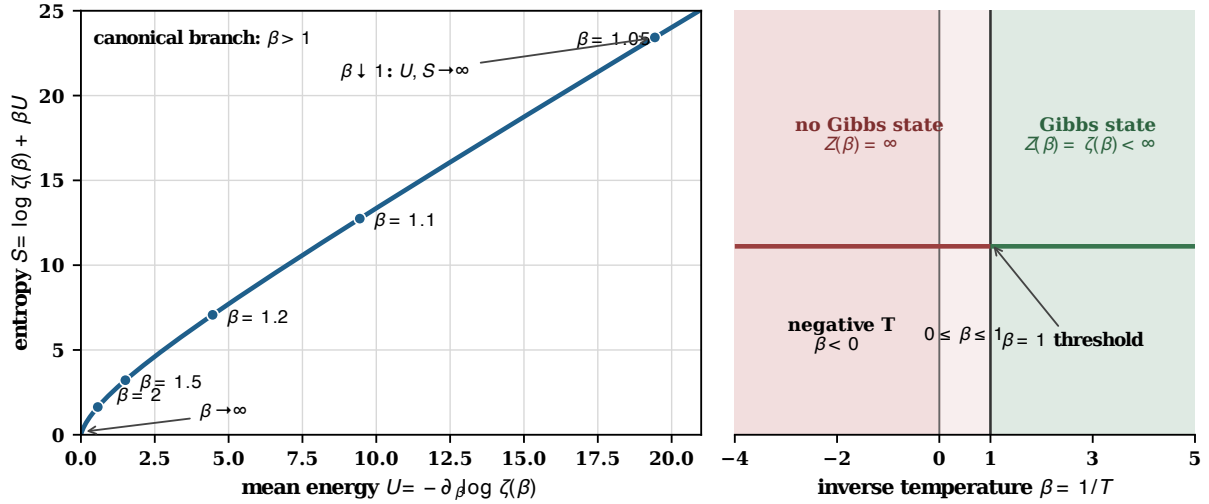


Figure 8: The canonical entropy–energy curve of the zeta Gibbs system and its inverse-temperature domain. Since the spectrum $E_n = \log n$ is unbounded above, $\sum_n e^{-\beta E_n}$ diverges for every $\beta \leq 1$. Thus the model has no normalized negative-temperature branch ($\beta < 0$); analytic continuation of ζ does not supply one.

10.1 Complex temperature and zeros

For $\beta > 1$, normalize the Gibbs law

$$\Pr_{\beta}(N = n) = \frac{n^{-\beta}}{\zeta(\beta)}.$$

Then

$$\frac{\zeta(\beta + it)}{\zeta(\beta)} = \mathbb{E}_{\beta}[e^{-it \log N}] \quad (12)$$

is the characteristic function of the random log-energy. A zero in this half-plane would be exact destructive phase cancellation, but the Euler product proves that no such zero exists.

The nontrivial zeros lie in $0 < \Re s < 1$, outside the normalizable Gibbs regime of (10). Analytic continuation assigns values there, but no positive probability measure $n^{-1/2}/\zeta(1/2)$ exists. Consequently

$$\rho = \frac{1}{2} + i\gamma$$

should not presently be called a literal complex energy level. A safer statement is that γ is a logarithmic Mellin frequency: terms such as $x^{\rho} = x^{1/2}e^{i\gamma \log x}$ oscillate in log scale in explicit formulas.

The quantum-chaos analogy reverses the roles. It seeks an unknown Hermitian Hamiltonian whose eigenvalues are the ordinates γ , while the prime powers p^k contribute periodic-orbit lengths $k \log p$. Berry and Keating's $H = xp$ program is a prominent version of this analogy [14]. It does not currently identify a zero with a quantum composition of the independent prime modes in (10).

10.2 What the completion adds

The completed function

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

has three conceptually different pieces.

1. $\zeta(s) = \prod_p(1 - p^{-s})^{-1}$ is the product of the finite-prime local factors.
2. $\pi^{-s/2}\Gamma(s/2)$ is the archimedean local factor, obtained as the Mellin transform of a Gaussian.
3. $s(s-1)/2$ cancels the poles at 0 and 1, producing an entire function.

If

$$\theta(t) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 t},$$

then for $\Re s > 1$

$$\int_0^{\infty} (\theta(t) - 1)t^{s/2} \frac{dt}{t} = 2\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Poisson summation gives $\theta(t) = t^{-1/2}\theta(1/t)$, and splitting the integral at $t = 1$ yields analytic continuation and the reflection

$$\xi(s) = \xi(1-s).$$

In Tate's adelic formulation, the finite and archimedean factors are local Mellin transforms and the functional equation is global Fourier duality [12]. For the Gibbs analogy, this is the main lesson of the completion: the global object is not only a product of positive local oscillators; it is completed by a self-dual archimedean mode and a scale reflection.

11 The Weil matrix E and the rate matrix M

For an admissible test profile g with Mellin transform $\widehat{g}$, the Weil quadratic form may be written schematically as

$$Q_W(g) = \sum_{\rho} \widehat{g}(\rho) \overline{\widehat{g}(1 - \bar{\rho})},$$

with the standard symmetric convergence. Polarization gives $B_W(g, h)$, and a finite family gives the Hermitian matrix

$$E_{ij} = B_W(g_i, g_j).$$

Weil’s explicit-formula criterion says that positivity for every admissible family is equivalent to the Riemann hypothesis [13]. Under RH, $1 - \bar{\rho} = \rho$, so E becomes a Gram matrix of the sampled Mellin transforms.

This matrix and the operational exchange matrix have different definitions:

	rate matrix M	Weil matrix E
input	maps and allowed processors	admissible Mellin test profiles
direction	directed	Hermitian
composition law	$M_{ij}M_{jk} \leq M_{ik}$	sesquilinear polarization
positivity notion	no profitable cycle	positive semidefinite quadratic form
arithmetic data	implementation counts	primes, gamma factor, and zeros
known bridge	none in this paper	none in this paper

One exploratory construction multiplies a map profile $\mathcal{Z}_i(s)$ by a log-Gaussian Mellin window and samples the first N known critical-line zeros. With $\sigma = 1/20$, omitting a common positive factor, the real matrix is

$$E_{ij}^{(N)} = 2 \sum_{n=1}^N e^{-\gamma_n^2/400} \operatorname{Re} \left(\overline{\mathcal{Z}_i(\rho_n)} \mathcal{Z}_j(\rho_n) \right).$$

For zeros already sampled on the critical line this is visibly a Gram matrix, so its positive semidefiniteness is automatic. For $N = 100$, the observed ranks are:

family	nodes	rank
quadratic ternary forms over $\mathbb{F}_3$	5	5
quadratic homogeneous $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3^2$	7	6
displayed quadratic representatives over $\mathbb{Q}_3$	8	5
displayed quadratic representatives over $\mathbb{Q}_2$	12	5

The rank losses come from exact profile identities or repeated local zeta profiles. They say nothing yet about affine conversion. In particular, positive semidefiniteness of these finite-height Gram matrices is not evidence that no-arbitrage implies Weil positivity, and it is not an independent test of RH.

To connect M and E , one would need at least:

1. a canonical assignment from polynomial-map resources to admissible adelic test profiles;
2. compatibility of that assignment with processor composition and Cartesian powers;

3. a theorem translating operational inequalities into the sesquilinear explicit formula, or conversely.

No such construction is presently known here.

12 What is classical, what is new, and what is open

The main contribution is a new angle and a reproducible synthesis, not a claim to have rediscovered quadratic-form theory. The following ledger states the current novelty boundary.

topic		prior status	contribution and claim level
left–right equivalence		classical in singularity and orbit theory	use the same processor equation simultaneously for equivalence, exact singular reachability, and block exchange
quadratic normal forms		classical Witt, discriminant, radical, and Arf theory	a uniform resource-oriented derivation of the nodes and arrow direction over finite, real, complex, and local fields
finite-field counts	quadratic	consequences of classical orbit–stabilizer calculations	closed formulas and a reproducible atlas of class-sized Hasse diagrams for all odd q , all even $q \geq 4$, and the exceptional $q = 2$ case; the exact compilation appears to be new, but not the ingredients
homogeneous and cubic cases		instances belong to invariant theory and finite computation	candidate new computational datasets: the $\mathbb{F}_3$ five-, seven-, fifty-, twenty-six-, and nineteen-class diagrams, and the 110-class generated preorder over $\mathbb{F}_8$
asymptotic rates		resource theories and asymptotic spectra are classical; the signature formula is proved in the companion paper	attach the solved signature rate to polynomial-map orbit nodes while explicitly separating it from the affine operational rate
cycle return		cycle means and max–plus algorithms are classical	apply them to map-class exchange matrices as round-trip-loss and observer-resolution diagnostics
zeta as partition function		Euler product and Bost–Connes are classical	place the prime-mode Gibbs picture next to map profiles and show exactly where the analogy ceases to be an ordinary Gibbs system
RH and Weil positivity		classical Weil criterion	no new RH claim; the missing M -to- E bridge is formulated as an open problem

Three items are the strongest candidates for genuinely new results after a full literature review:

1. the explicit all- q affine-processor Hasse atlas for degree-at-most-two maps $\mathbb{F}_q^2 \rightarrow \mathbb{F}_q$, with orbit sizes at every node;

2. the exhaustive homogeneous and cubic finite-field datasets and their verification programs;
3. the combined study of the exact degeneration poset, the solved fiber-signature exchange observer, and the max-times cycle losses on the same named classes.

These should be advertised as new computations and organization unless a prior source is found. The no-arbitrage proof itself is elementary and should not be sold as a deep new theorem; its value is that it places a precise constraint on every rate matrix generated by the framework.

13 Open problems

Question 13.1 (Affine coding theorem). Can $C_{\text{aff}}(g \rightarrow f)$ be characterized by a complete family of algebraic or analytic monotones, analogous to the partition functions that solve the finite-signature theory?

Question 13.2 (The seven 2-adic anisotropic classes). Are distinct anisotropic binary forms over $\mathbb{Q}_2$ asymptotically interconvertible at a rate strictly greater than $1/2$? Does any pair have rate one without a finite rate-one conversion?

Question 13.3 (Cycle separation). For which processor categories does a uniform gap $r(\mathcal{C}) \leq 1 - \varepsilon$ over all nontrivial cycles characterize an asymptotic separation of equivalence classes? When does a cycle mean of one imply asymptotic reversibility rather than merely a coarse observer?

Question 13.4 (Functorial zeta profiles). Is there a lattice-independent or adelic profile assignment that is monotone under affine implementation, multiplicative under Cartesian products, and rich enough to bound or determine C_{aff} ?

Question 13.5 (The M -to- E bridge). Can a natural completion of polynomial-map resources produce admissible Weil test functions in a way that turns no-arbitrage or conversion monotonicity into positivity of a Hermitian form? A successful answer must explain why a directed max-times matrix should control a sesquilinear matrix; superficial entrywise transformations are insufficient.

14 Conclusion

The equation $f = h_{\text{out}} \circ g \circ h_{\text{in}}$ is deliberately small. With invertible affine processors it gives classical equivalence classes. With singular affine processors it gives a degeneration preorder and Hasse diagrams. With Cartesian powers it gives exchange rates, directed triangle inequalities, and no-arbitrage cycle bounds. With the fiber signature observer it gives partition functions and Gibbs curves already solved in the companion paper.

Quadratic maps show the benefit of the framework because the answer is recognizable: rank, radical, isotropy, discriminant, and the Arf/trace invariant reappear as resource monotones and branching rules. The same language continues from finite fields to real, complex, and p -adic analytic maps, and from scalar quadratics to homogeneous tensors and cubics.

The number-theoretic continuation is promising but incomplete. The Euler product really is a joint partition function of prime occupation modes for $\Re s > 1$; the completed zeta function really is assembled from finite and archimedean local factors and Fourier–Mellin duality. But the nontrivial zeros are not ordinary Gibbs energy levels, and the operational matrix M is not the Weil matrix E . The present contribution is to make the common language precise enough that a future bridge, if it exists, will have a clear mathematical statement.

A A visual atlas over finite fields

The formulas in Section 4 say that all odd finite fields share one six-node shape, while all even fields of order at least four share one seven-node shape. The diagrams below retain the shape and change the orbit sizes. They are included because the class-size labels make the orbit formulas concrete, and because the exceptional $\mathbb{F}_2$ collapse is easiest to remember visually.

B Two cubic processor choices

The next pair shows why the processor class is part of the theorem. The underlying universe is the same set of 6,561 cubic polynomial functions $\mathbb{F}_3^2 \rightarrow \mathbb{F}_3$. Affine input processors retain fourteen classes. The preorder generated by quadratic input processors collapses them to three.

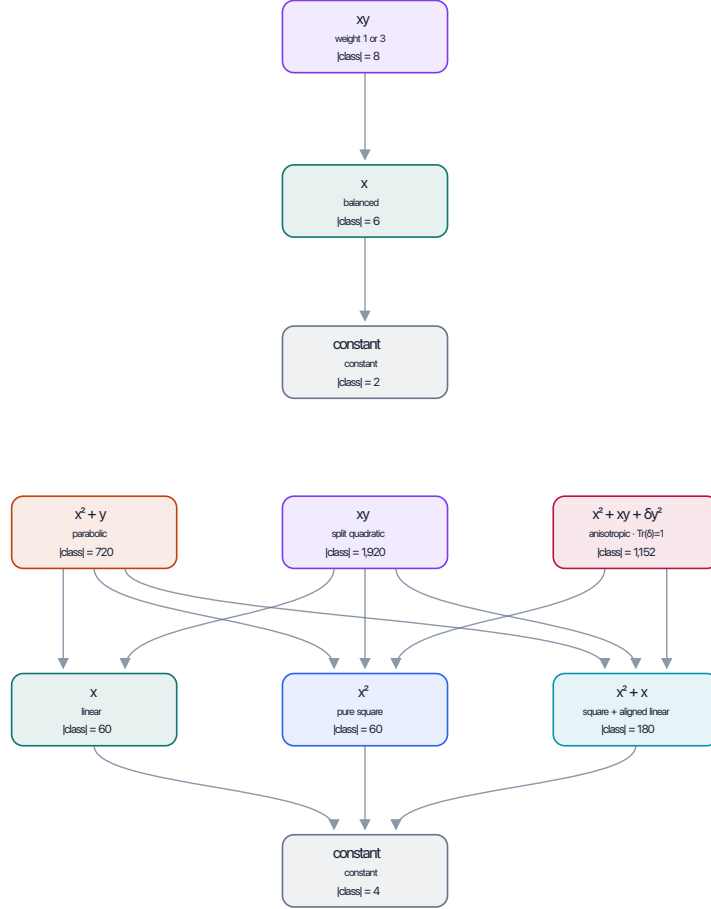


Figure 9: The exceptional three-class, two-cover function poset over $\mathbb{F}_2$, followed by the generic seven-class, eleven-cover poset over $\mathbb{F}_4$. Both use possibly singular affine processors.

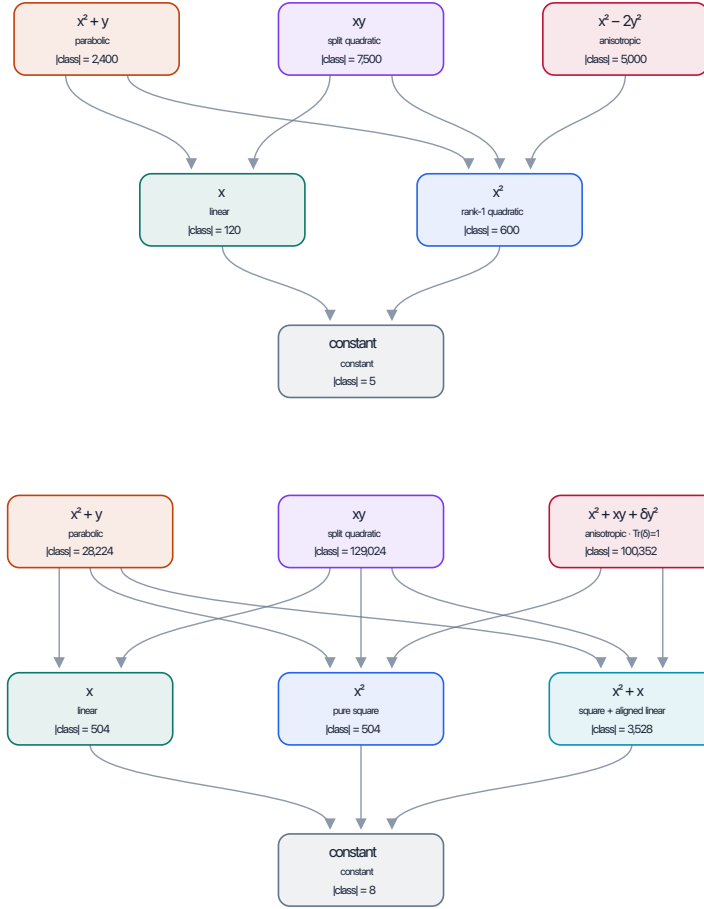


Figure 10: Quadratic maps with singular affine processors over $\mathbb{F}_5$ (six classes, seven covers) and $\mathbb{F}_8$ (seven classes, eleven covers). The field parity changes both the normal forms and the poset shape.

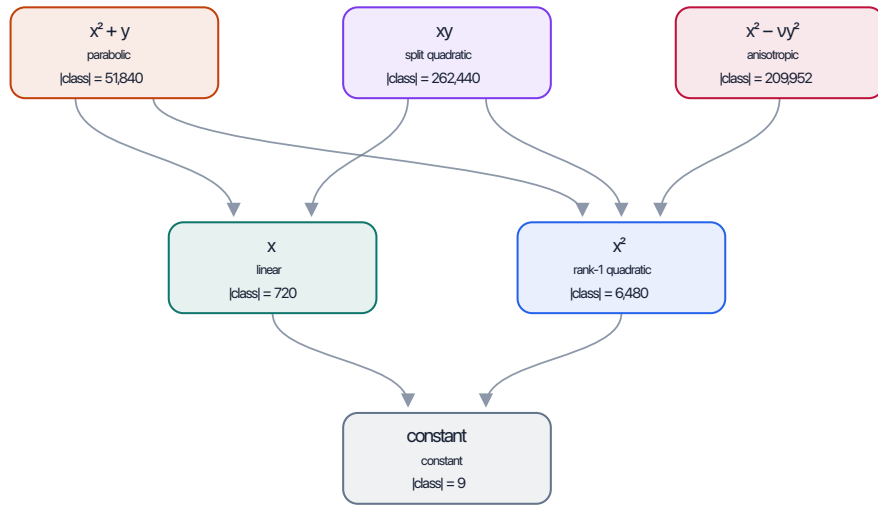


Figure 11: Quadratic maps over $\mathbb{F}_9$ with singular affine processors: six classes and seven covers. The node labels record the field-dependent orbit sizes.

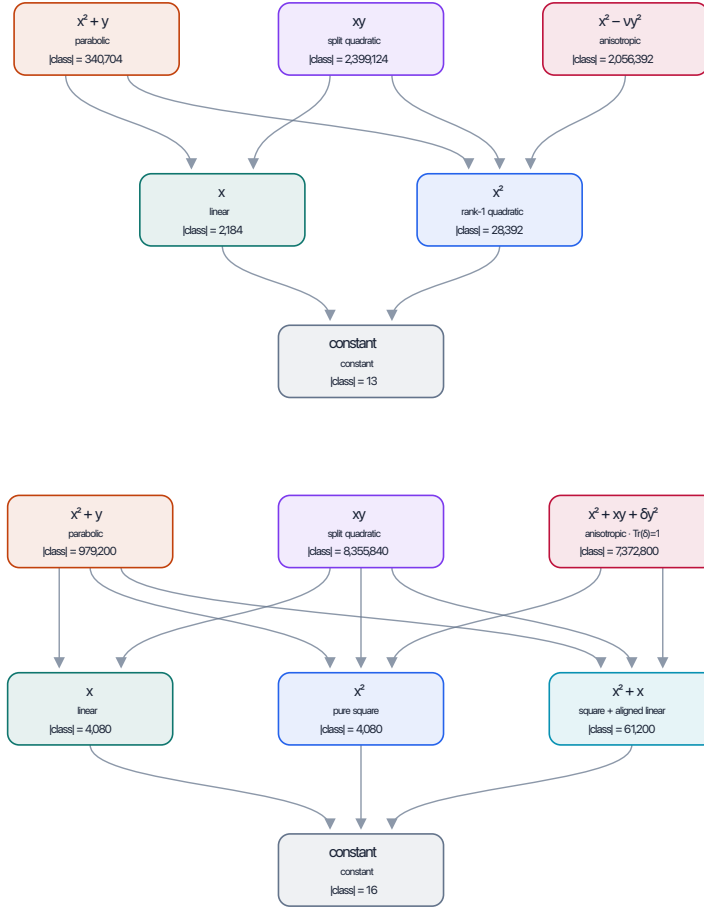


Figure 12: Quadratic maps with singular affine processors over $\mathbb{F}_{13}$ (six classes, seven covers) and $\mathbb{F}_{16}$ (seven classes, eleven covers).

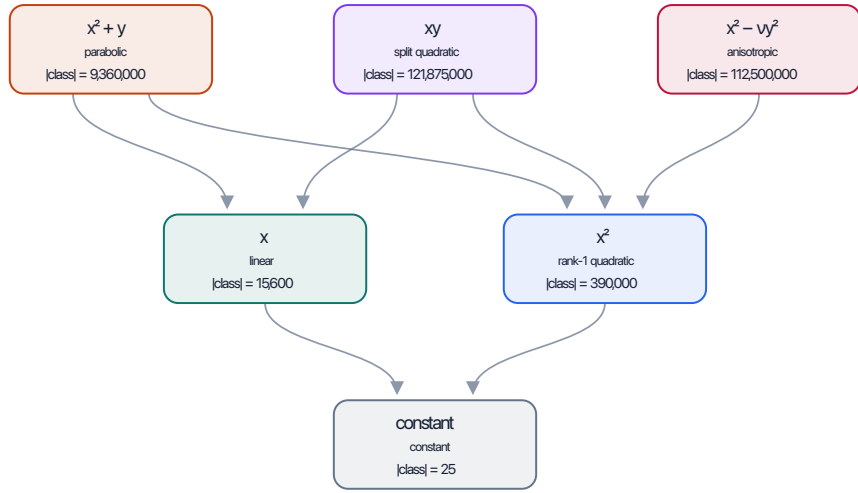


Figure 13: The odd-field diagram over $\mathbb{F}_{25}$, containing $25^6 = 244,140,625$ coefficient vectors in six affine classes and seven covers; singular affine processors define the order.

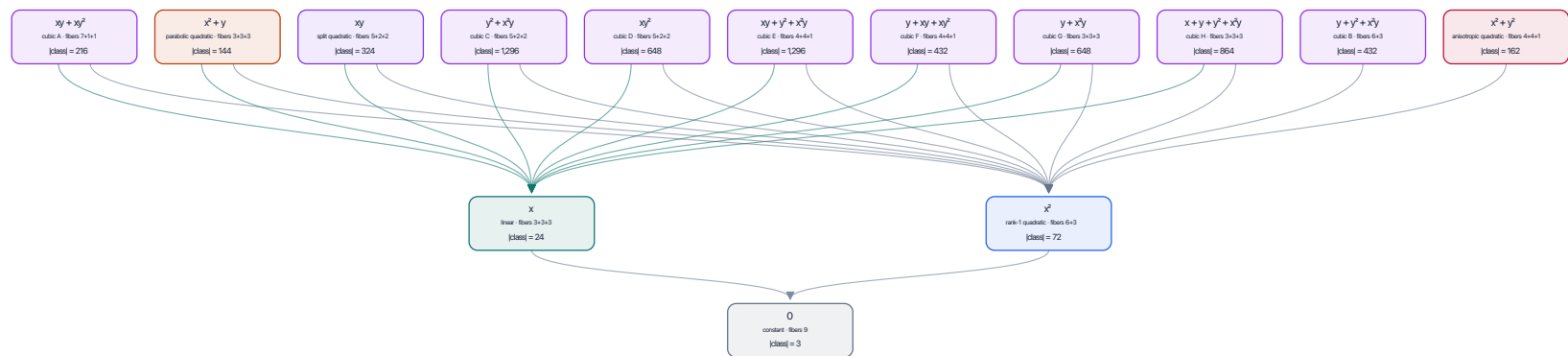


Figure 14: Cubic functions over $\mathbb{F}_3$ with affine input and output processors: fourteen equivalence classes and twenty-two Hasse covers.



Figure 15: The generated preorder with quadratic input and linear output processors: three mutual-reachability classes and two covers. It is not a group-orbit classification by invertible quadratic substitutions.

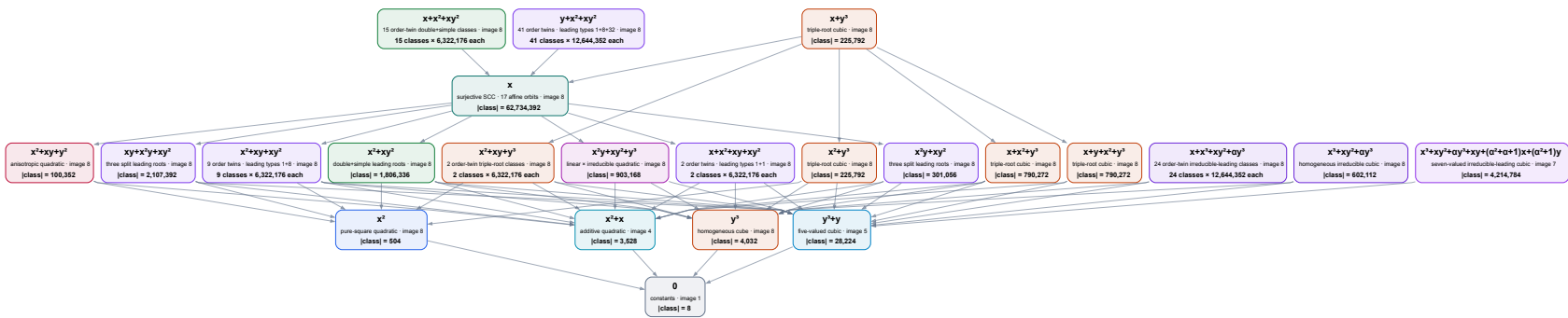


Figure 16: The corresponding quadratic-input calculation over $\mathbb{F}_8$: 110 classes grouped into 23 boxes of order-theoretic twins, with 57 covers between the boxes. Output processors are linear.

Code and data availability

The diagrams, exhaustive classifiers, residue calculations, exchange-rate library, and cycle tables are contained in the accompanying repository. The main regeneration commands are:

- `cli/finite_field_map_poset_cli` for the affine quadratic atlas;
- `cli/homogeneous_tensor_poset_cli` for the homogeneous cases;
- `cli/cubic_map_poset_cli` for the cubic processor comparisons;
- `analysis/polynomial_map_exchange_examples.py` for the exchange matrices and cycle means used here.

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